[Team Name]

CODESLAYER 2K26

DevSphereIndia

Project Title : **TakeSure — Confidence before taking something.** Ayurveda × Allopathy herb–drug interaction checker: photo of prescription + photo of product → sourced, plain-language safety verdict + Ministry of Ayush ADR report.

Team Name : [Team Name]

Team Members
(Names + Roles Briefly):

[Member 1] — Team lead · backend, rule engine & pitch
[Member 2] — Frontend · web app, Hindi/English UI, result card
[Member 3] — Medical research · knowledge base & evidence grading
[Member 4] — OCR, PDF/ADR export & demo video

Tracks Chosen: **Healthcare** · **AI/ML** · **Open Innovation**

Project explanation video (Google Drive): [paste link here — set to "Anyone with the link can view"]

Aligned with the Ministry of Ayush Pharmacovigilance Programme for ASU&H drugs · decision support, never diagnosis

PROBLEM STATEMENT & TARGET AUDIENCE

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Problem We're Solving

- **India runs two medical systems side by side — and nobody checks the combination.** A thyroid patient on Thyronorm adds ashwagandha for stress; a transplant patient on tacrolimus drinks giloy kadha; a cardiac patient on warfarin takes guggul for cholesterol. Ashwagandha can raise thyroid hormone, giloy can oppose immunosuppression, guggul can increase bleeding.
- The allopathic doctor rarely asks, the patient rarely tells, the chemist has no tool, and the product label says nothing about allopathic medicines.
- Existing checkers (Medscape, Drugs.com, 1mg) only know allopathic × allopathic pairs; Western herb databases are English-only, paid and do not know Indian brands like Himalaya Ashvagandha or Dabur Giloy Ghanvati.

37%	65%	19%	1,473
of hospital patients used allopathic + herbal medicines together	of herbal users lacked knowledge of dose, side effects, interactions	of Indian patients disclose herbal use to their doctor	ASU&H adverse-event reports nationally since 2018

Impacted: patients on thyroid, diabetes, BP/heart, blood-thinner, transplant or epilepsy medicines and their families; doctors who never hear about the co-use; the Ministry's pharmacovigilance network that never receives the reports.

Target Users

B2C · Patients & caregivers

Adults 30–70 on long-term medicines who also take Ayurvedic products; family members managing an elder's medicines; Hindi or English, phone-first.

B2B · Clinicians & pharmacies

MBBS/MD doctors, BAMS vaidyas, hospital pharmacies, retail chemists, telemedicine and hospital-HIS platforms (API).

B2G · Regulators

Ministry of Ayush / NPvCC at AIIA and the 99 Peripheral Pharmacovigilance Centres: structured ICSRs and co-use analytics.

OUR UNIQUE SOLUTION

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Key Features

1. **Scan —** Photograph the prescription and the Ayurvedic product label (or type names). OCR reads them; every name is shown for confirmation before anything is checked.

2. **Match —** Indian brand → generic (Thyronorm → levothyroxine), product → herbs (Chyawanprash → amla, ashwagandha…) via a curated bridge dictionary + fuzzy matching.

3. **Decide —** Deterministic rule engine over a cited knowledge base → GREEN / AMBER / RED / NOT IN DATABASE, with timing gaps (e.g. keep 2–3 h between Triphala and oral medicines).

4. **Explain —** AI rewrites only the matched evidence into simple Hindi/English + 3 questions to ask the doctor. It cannot add, remove or re-grade an interaction.

5. **Report —** One-page PDF with sources for the doctor; one-tap suspected-ADR report in the Ministry of Ayush format for the Ayush Suraksha portal.

Why It's Different

	Ayurveda × allopathy	Indian brands & Hindi	Sources & evidence grade	ADR to Ministry	Score
Medscape · Drugs.com · 1mg	★☆☆☆☆	★★☆☆☆	★★★★☆	☆☆☆☆☆	7 / 20
MSKCC · Natural Medicines	★★★☆☆	★☆☆☆☆	★★★★★	☆☆☆☆☆	9 / 20
TakeSure	★★★★★	★★★★★	★★★★☆	★★★★★	**19 / 20**

★ = our assessment of each tool against the five things an Indian patient needs; sources: product sites & documentation.

Novelty

- **"Verdict from published data; AI only explains."** The risk level never comes from a language model — it comes from a versioned, cited knowledge-base row. Turn the AI off and the verdict is identical.
- **The missing bridge table:** Indian brand names × Ayurvedic herb synonyms (Sanskrit/Hindi/Latin) × graded published evidence — open, auditable, co-curated with Ayurveda colleges.

concept of a result card

! AMBER — Ashwagandha + Levothyroxine · Evidence B · recheck TSH in 6–8 weeks · ask your doctor

TECH STACK + ARCHITECTURE

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Tech Stack

Frontend: Next.js (React) + TypeScript, Tailwind CSS, Framer Motion · installable PWA · Hindi/English UI

Backend: FastAPI (Python), Pydantic v2 validation, SQLAlchemy · REST API with typed errors

Database: PostgreSQL (Supabase) in production; SQLite for offline/demo · tables: herbs, drugs, products, interactions, sources

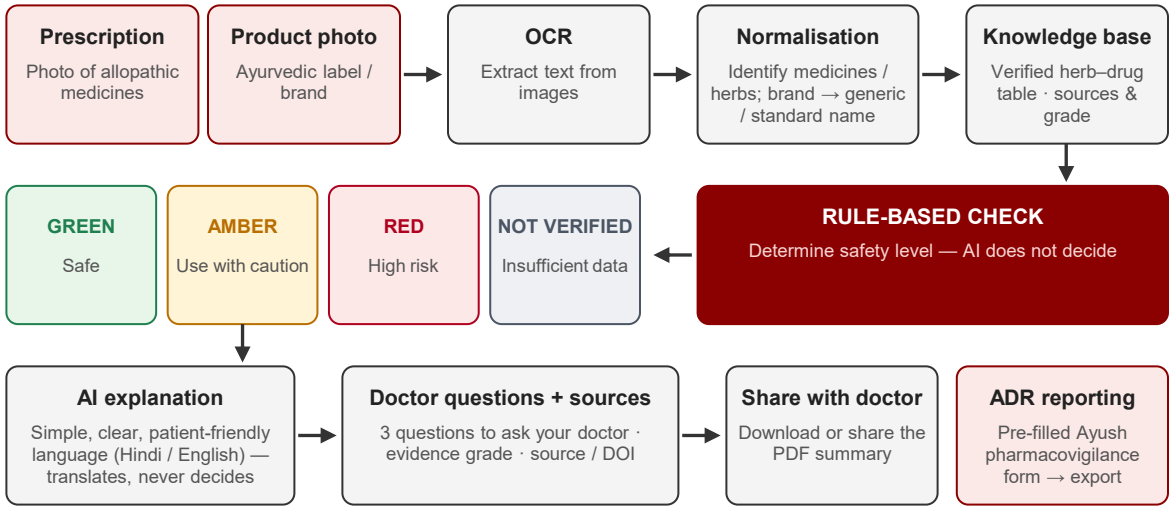
OCR: Tesseract (eng + hin) on the server; Google Vision / Groq vision model as optional cloud reader

Matching: RapidFuzz token-set matching; ≥85 accept, 70–85 ask the user, <70 not recognised

AI explainer: Groq API (Llama / GPT-OSS) with JSON-schema output, temperature 0, facts-only prompt, fallback text

PDF & hosting: ReportLab for doctor summary + ADR form · Vercel (web) + Render/Railway (API)

Architecture — project workflow



Decision support only · never diagnosis · always ask your doctor

Verdict path: photos → OCR → names confirmed by the user → knowledge-base lookup → rule engine (worst row wins: RED > AMBER > GREEN; unknown is never GREEN). The AI receives only the matched rows and rewrites them; invalid or directive output falls back to knowledge-base text. Uploads are deleted after extraction.

FEASIBILITY AND SHOWSTOPPERS

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Feasibility

- **All components are proven and off-the-shelf:** OCR (Tesseract/Vision), fuzzy matching (RapidFuzz), relational lookup (PostgreSQL), LLM rephrasing (Groq). No model training — the MVP is a data + rules product.
- **Resources:** 4 members with clear roles — backend & rule engine, frontend, medical research/knowledge base, OCR & ADR/PDF; laptops run the full stack locally (SQLite), so no paid cloud is needed on-site.
- **Research before the hackathon, code during it:** a knowledge base of 40–60 high-frequency herb × drug pairs (15 herbs × 12 drug classes) curated from PubMed, LiverTox, MSKCC, WHO monographs and the Ayurvedic Pharmacopoeia; every row has a source and an A/B/C grade.

36-hour build plan

0–2 h	2–10 h	10–16 h	20–28 h	28–32 h
Setup, schema, load KB	Core loop: photo → OCR → confirm → verdict	Explainer, ADR form, PDFs	Hindi UI, result card polish, dashboard	Freeze, test on real boxes, rehearse

Showstoppers → Mitigation

Risk / blocker	Mitigation
OCR misreads Indian handwriting or faded labels	Confirm-names screen before any lookup; manual entry always available; brand dictionary catches partial reads
Herb or drug not in our database	Say “Not in database” — never guess, never show GREEN; log the gap as a curation task
LLM hallucinates medical advice	Model receives only matched rows; JSON-schema output; “stop/start/dose” language rejected; deterministic fallback text; verdict unchanged with AI off
Weak evidence for many herbs	Every row shows grade A/B/C and a DOI/PMID; grade-C evidence caps at AMBER; weak is shown as weak
Venue internet or API outage	Whole stack runs offline on a laptop (SQLite + Tesseract); cached explanations as backup
Liability / medical scope	Decision support only: “ask your doctor”, never “stop the medicine”; no diagnosis, no dosing; disclaimer on every screen and PDF

USP(UNIQUE SELLING PRICE) & BUSINESS MODEL

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USP (UNIQUE SELLING PRICE)

- **India's first safety layer between the two medical systems people already mix** — nothing public today checks an Ayurvedic product against an allopathic prescription.
- **Verdict from published data; AI only explains** — auditable rules over cited rows; the opposite of a “ChatGPT wrapper”.
- **Built for India:** Indian brand names, Sanskrit/Hindi synonyms, Hindi explanations, Ministry of Ayush ADR format, Ayush Suraksha portal link.
- **Honest by design:** evidence grades, timing gaps instead of alarms, “not in database” instead of guesses.

BUSINESS MODEL

Free for citizens (B2C) — Web/PWA checker in Hindi & English; grows the knowledge base and the ADR signal.

API & counter tool (B2B) — Per-query / subscription API for pharmacy chains, hospital HIS, telemedicine apps; QR on the chemist's bill.

Programmes & grants (B2G / CSR) — Co-use analytics and structured ICSRs for the Ministry of Ayush pharmacovigilance network; CSR funding from Ayurvedic brands that want a trust layer.

Platform — Ayush Grid / ABDM integration — verdict attached to the ABHA health record; open KB co-curated with Ayurveda colleges.

References

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- Ministry of Ayush: Pharmacovigilance Programme for ASU&H drugs (AOGUSY); NPvCC at AIIA; ICSR figures (Lok Sabha reply).
- Ayush Suraksha Portal (May 2025, SC order WP(C) 645/2022) — ayushsuraksha.com.
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- Stockley's Herbal Medicines Interactions, 2nd ed. (2013) — evidence-grading model; Ayurvedic Pharmacopoeia of India (PCIM&H).
- Docs: Tesseract, RapidFuzz, FastAPI, Next.js, PostgreSQL, ReportLab, Groq · comparators: Medscape, Drugs.com, 1mg.